FENOC™
FirstEnergy Nuclear Operating Company

Perry Nuclear Power Plant
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David B. Hamilton
Vice President

440-280-5382

October 4, 2017
L-17-304

10CFR50.73(a)(2)(v)(D)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2017-004, "Loss of Safety Function for High Pressure Core Spray Suppression Pool Level Instrumentation". There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Nicola Conicella, Manager – Regulatory Compliance, at (440) 280-5415.

Sincerely,

David B. Hamilton
Vice President

Enclosure:
LER 2017-004

cc: NRC Project Manager
NRC Resident Inspector
NRC Region III Regional Administrator

Enclosure
L-17-304

LER 2017-004

NRC FORM 366 (04-2017) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 3/31/2020

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Perry Nuclear Power Plant	05000-440	1 OF 3

4. TITLE: Loss of Safety Function for High Pressure Core Spray Suppression Pool Level Instrumentation

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
08	08	2017	2017 -	004 -	00	10	4	2017		05000
									FACILITY NAME	DOCKET NUMBER 05000

9. OPERATING MODE: 1

10. POWER LEVEL: 100

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT:	TELEPHONE NUMBER (Include Area Code)
Tony Kledzik – Regulatory Compliance	440-280-6188

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
D	BG	LIT	X999	Y					

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On August 8, 2017, at 1554 hours, while the plant was at 100 percent rated thermal power, during restoration from testing of the High Pressure Core Spray (HPCS) Suppression Pool (SP) Level High Instrumentation, unexpected as-left indications were found that impacted both of the required channels of instrumentation. With both SP level instruments inoperable, a loss of safety function existed.

While venting the sensing line, the HPCS system was aligned to the suppression pool water source. This source of water is HPCS's safety-related source of water. The automatic suction swap on high suppression pool level is implicitly assumed in the accident and transient analysis since it assumes that the HPCS suction source is the suppression pool. Since the HPCS system was aligned to the suppression pool when the failure occurred, the assumptions of the accident analysis are met, and no safety system functional failure occurred.

The cause for the unexpected as-left indications is due to air entrained in the sensing line which came out of solution.

The safety significance of this event is considered to be small. This event is being reported under 50.73(a)(2)(v)(D) for a loss of safety function.

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		YEAR	SEQUENTIAL NUMBER	REV NO.
Perry Nuclear Power Plant Unit 1	05000-440	2017	- 004	- 00

NARRATIVE

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

INTRODUCTION

On August 8, 2017, at 1554 hours, while the plant was at 100 percent rated thermal power, during restoration from testing of the High Pressure Core Spray (HPCS) [BG] Suppression Pool (SP) [BT] Level High Instrumentation, [LIT] unexpected as-left indications were found that impacted both of the required channels of instrumentation. With both SP level instruments inoperable, a loss of safety function existed.

EVENT DESCRIPTION

On August 8, 2017, at 1321 hours, surveillance testing commenced on the HPCS SP level high instrumentation and one channel was declared inoperable. On August 8, 2017, at 1554 hours, during restoration from surveillance testing of the HPCS SP level high instrumentation, unexpected as-left indications were found that impacted both of the required channels of instrumentation. Technical Specification (TS) 3.3.5.1 "Emergency Core Cooling System (ECCS) Instrumentation" condition D was entered. Condition D requires HPCS to be aligned to the SP; however, HPCS was already aligned to the SP since earlier in the day in anticipation of the testing.

On August 8, 2017, at 1635 hours, venting was performed on the level instrumentation sensing line and the erroneous reading was corrected. The restoration from the venting was completed and the level instrumentation was declared operable on August 8, 2017, at 1747 hours.

On August 8, 2017, at 2022 hours, event notification EN# 52891 was made to the NRC Operations Center to report a loss of safety function.

Suppression Pool Water Level High – TS 3.3.5.1

SP Water Level-High signals are initiated from two level transmitters. The logic is arranged such that either transmitter and associated trip unit can cause the suppression pool suction valve to open and the Condensate Storage Tank (CST) [KA] suction valve to close. This function is implicitly assumed in the accident and transient analyses, which take credit for HPCS, since the analyses assume that the HPCS suction source is the suppression pool.

CAUSE

The cause for the unexpected as-left indications that impacted both of the required channels of instrumentation is related to air entrained in the sensing line caused by air being introduced from air accumulation in the suppression pool over time which moved to a vertical section of the sensing line during venting.

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NARRATIVE

EVENT ANALYSIS

While venting the instrumentation, the HPCS system was aligned to the suppression pool water source. This source of water is HPCS's safety-related source of water. The automatic suction swap on high suppression pool level is implicitly assumed in the accident and transient analysis since it assumes that the HPCS suction source is the suppression pool. Since the HPCS system was aligned to the suppression pool when the failure occurred, the assumptions of the accident analysis are met, and no safety system functional failure occurred.

Probabilistic Risk Assessment review concludes that the associated event, from a qualitative analysis, is of low risk (small safety) significance. The condition was introduced during the surveillance activity, was identified in a timely manner, and was immediately corrected. Additionally, HPCS was aligned to the inventory source that would have been the inventory source that would have been transferred to given the subject instrumentation functioned appropriately. The function/availability of HPCS was never affected. On this basis, there would be no corresponding change (delta) in core damage frequency (CDF), and no corresponding change (delta) in the large early release frequency (LERF). The delta CDF and delta LERF values would therefore be well below the acceptable thresholds of 1.0E-06/yr and 1.0E-07/yr, respectively, as discussed in Regulatory Guide 1.174. Therefore, the risk of this event is considered small in accordance with the regulatory guidance.

CORRECTIVE ACTIONS

Corrective actions include revising local leak rate testing instruction of the sensing line penetration to perform a fill and vent of the sensing line during every performance to reduce the buildup of entrained air.

PREVIOUS SIMILAR EVENTS

A review of LERs and the corrective action database for the past three years identified no similar events.

COMMITMENTS

There are no regulatory commitments contained in this report. Actions described in this document represent intended or planned actions, are described for the NRC's information, and are not regulatory commitments.